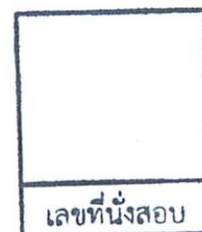


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2015

Subject: 394173 Mathematics III

Section: 15-16

Date: 02 December 2015

Time: 13.00-16.00

Name _____ ID _____ Class _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks, dictionary and Calculators are **not** allowed.
4. This is a closed book examination.
5. Electronic communication device are **not** allow in the examination room.
6. The examination has 8 pages (including this page), 8 questions and 1 part.
The total score is 35 points.
7. Write all your answers on this examination sheet.

No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	Total score

1. Assume that a, b, c are the real numbers such that $a, 2b, 3c$ is the geometric sequence and $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ is the arithmetic sequence. Find the sum of $\frac{a}{c} + \frac{c}{a}$?
(4 points)

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2. Find each limit, or state that the limit does not exist and explain your reasoning.

2.1 $a_n = \sqrt{n^6 + 5} - n^3$ (2 points)

2.2 $a_n = (-1)^{n+3} \frac{\sqrt[3]{8n^3 - 3n + 7}}{\sqrt[4]{1 + 5n^2 + 2n^4}}$ (2 points)

3. Find the sum of first ten terms of $2 \cdot 1 + 5 \cdot 4 + 8 \cdot 9 + 11 \cdot 16 + \dots$ (Use $\sum$)

(5 points)

4. In the geometric series, if $a_1 = 3$, $a_n = 96$ and the sum of first n terms is 189.
Determine n and r . (4 points)

5. Find the sum of the infinite series

$$\frac{3}{1} + \frac{3}{1+2} + \frac{3}{1+2+3} + \frac{3}{1+2+3+4} + \dots$$

(5 points)

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6. Assume that $\vec{u} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{v} = -\hat{i} + \hat{j} + 3\hat{k}$. Find the vector has length as $|\vec{u} \cdot \vec{v}|$ which has the same direction as $\vec{u} - 2\vec{v}$? (4 points)

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7. Assume that $\vec{u}$, $\vec{v}$ and $\vec{w}$ are the vectors such that $\vec{u} + \vec{v} + \vec{w} = \vec{0}$ and $|\vec{u}| = 1$, $|\vec{v}| = 4$, $|\vec{w}| = 5$. Find $3|\vec{u} - \vec{v}| - 2|\vec{v} - \vec{w}|$? (5 points)

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8. Let $A(2, 0, 3)$, $B(2, 4, 5)$ and $C(1, x, 3)$ be the points, where x is a positive integer such that the area of triangle $\triangle ABC = 3 \text{ units}^2$. Find $\vec{BC} \cdot \vec{AC}$? (4 points)

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